

Pragati Bharati National Assessment Examination

1. Which of the following data structures operates on a Last-In, First-Out (LIFO) principle?

- (A) Queue
- (B) Stack
- (C) Binary Tree
- (D) Hash Map

2. What is the time complexity of searching for an element in a balanced binary search tree with n nodes?

- (A) $O(1)$
- (B) $O(n)$
- (C) $O(\log n)$
- (D) $O(n \log n)$

3. In relational database management systems, which normal form eliminates transitive dependencies?
- (A) First Normal Form (1NF)
 - (B) Second Normal Form (2NF)
 - (C) Third Normal Form (3NF)
 - (D) Boyce-Codd Normal Form (BCNF)

=== ANSWER KEY ===

- 1. B
- 2. C
- 3. C